

Saikat Das

AI/ML & Computer Vision • Edge AI • Backend Development

✉ saikatdasmoin47@gmail.com ✉ saikat.cse.20210104158@aust.edu

☎ +880 1790 070535

github.com/saikatdas47 linkedin.com/in/saikatdas47 Codeforces



PROFILE

Computer Science graduate specializing in AI/ML research, computer vision, and backend development, with a first-author journal paper under review at PLOS ONE. Hands-on experience in efficient deep learning, edge AI, and vision-language systems, plus building and deploying scalable Node.js services and data pipelines on AWS, Vercel, and Firebase. Strong grounding in data structures, algorithms, and system design, backed by 200+ solved competitive programming problems.

Research Interests: Computer Vision, Health AI, Efficient Deep Learning, Edge AI / TinyML, Vision Transformers, Vision-Language Models, NLP, Model Quantization & Compression, Hardware-Aware NAS.

TECHNICAL SKILLS

Languages	C++, Python, MATLAB, JavaScript, Java (basic), PHP (basic), SQL
AI / ML / DL	PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Hugging Face Transformers, Pandas, NumPy, NLP
Edge AI & Optimization	Quantization (TensorFlow Lite, ONNX), Pruning, Hardware-aware NAS, ARM Cortex deployment
Backend	Node.js, Express.js, REST APIs, Socket.io, JWT Auth, MVC, ETL Pipelines
Databases & Cloud	MongoDB, MySQL, Firebase, MongoDB Atlas, AWS, Cloudinary
Frontend	React.js (Vite), Tailwind CSS, EJS, Bootstrap
Tools & Concepts	Git/GitHub, Docker (basic), Linux, Jupyter, Google Colab, OS, Networks, System Design, OOP, DSA

PUBLICATIONS UNDER REVIEW

Saikat Das, et al. "A Deployment and Resource Aware Edge AI Framework for Real-Time Waste Classification with Biometric Authentication and Secure Logging"

PLOS ONE — Under review, submitted April 2026

PROJECTS

PingMe — Real-Time Chat Application Repo

Jan 2026

Node.js, Express, MongoDB, Socket.io, JWT, React (Vite), Tailwind CSS, Cloudinary

- Built and deployed a full-stack MERN chat app with real-time messaging over WebSockets (Socket.io).
- Implemented JWT authentication with bcrypt password hashing and protected REST APIs.
- Integrated Cloudinary image uploads; deployed frontend on Vercel and backend on Koyeb.

BlogApp — Backend-Focused Blog Platform Repo

Jan 2026

Node.js, Express, MongoDB Atlas, EJS, AWS

- Developed a production-style backend with full CRUD, middleware-based authentication, and protected routes.
- Applied MVC architecture with server-side rendering (EJS); deployed on AWS with a cloud database.

SmartWasteBin — Edge AI Waste Management Repo

Sep 2025

CNN, Edge AI, IoT, Blockchain, Firebase — led to a journal submission (*PLOS ONE*)

- Quantization-aware training and architecture search to optimize latency/accuracy trade-offs on ARM Cortex-M devices.
- Blockchain-based immutable logging for audit trails and Firebase cloud sync for real-time monitoring.
- Biometric authentication integration for secure operator access.

SentimentSphere — Sentiment Analysis Pipeline Repo

Apr 2025

Python, Scikit-learn, NLP, BeautifulSoup

- Built an end-to-end sentiment classifier using Logistic Regression with TF-IDF and custom preprocessing (stopword removal, stemming).
- Scraped **50k+ reviews** with BeautifulSoup; achieved **89% F1-score**.
- Explored model calibration and bias analysis for production deployment.

Paira — Messenger Application Repo

Jan 2025

Java, Socket Programming, Firebase

- Designed a real-time chat app with user authentication and secure Firebase media storage.

Hotel Management System Repo

Jul 2024

PHP, MySQL, Bootstrap

- Developed a dynamic hotel booking platform with secure authentication and reservation management.

EDUCATION

B.Sc. in Computer Science and Engineering

Graduated 2026

Ahsanullah University of Science and Technology (AUST)

CGPA: 3.32 / 4.00 • **3.65 in last 40 credits**

HSC — Chittagong College • GPA 5.00/5.00

2020

SSC — Saint Placid's High School • GPA 5.00/5.00

2018

ACHIEVEMENTS

- Solved **200+ problems** on competitive programming platforms (Codeforces, LeetCode).
- Regular participant in ICPC-style regionals and intra-university programming contests.
- Reviewer activity for undergraduate research symposium (AUST CSE Fest 2024).
- Completed graduate-level coursework in Advanced Machine Learning and Digital Image Processing.

CERTIFICATIONS

MATLAB Onramp — MathWorks	Jul 2026	(Certificate) (Report)
Machine Learning Onramp — MathWorks	Jul 2026	(Certificate) (Report)
Deep Learning Onramp — MathWorks	Jul 2026	(Certificate) (Report)
Introduction to Data Engineering — IBM	Feb 2026	(Certificate)
Vision Language Models (VLM) Bootcamp — OpenCV University	Feb 2026	(Certificate)
TensorFlow-Keras Bootcamp — OpenCV University	Dec 2025	(Certificate)
PyTorch Bootcamp — OpenCV University	Dec 2025	(Certificate)
Python For Beginners — OpenCV University	Dec 2025	(Certificate)
Prompt Design in Vertex AI — Google Cloud	Jan 2026	(Badge)
Responsible AI: Applying AI Principles — Google Cloud	Jan 2026	(Badge)
Professional ML Engineer Study Guide — Google Cloud	Jun 2025	(Badge)
Introduction to Large Language Models — Google Cloud	Jul 2025	(Badge)
Introduction to Responsible AI — Google Cloud	Jul 2025	(Badge)
Introduction to Generative AI — Google Cloud	Jun 2025	(Badge)
Professional Networking for Career Growth — HP LIFE / Mission 44	Jan 2026	(Certificate)
Build a Free Website with WordPress — Coursera	Feb 2026	(Certificate)

ADDITIONAL INFORMATION

Languages: English (Fluent), Bengali (Native)

Strengths: Analytical problem-solving, technical writing, cross-disciplinary collaboration, rapid prototyping

Interests: AI for sustainability, efficient deep learning, multimodal reasoning, open-source scientific computing

REFERENCES

Ms. Nusrat Jahan — Thesis Advisor, Research Mentor
Assistant Professor, Dept. of CSE, AUST
✉ nusratjahan.cse@aust.edu

Mr. Mustofa Ahmed — Co-Supervisor
Lecturer, Dept. of CSE, AUST
✉ mustofa.cse@aust.edu